18 Write $\frac{3(x-1)}{x^2-4x-5} - \frac{2}{x-5}$ as a single fraction in its simplest form.

	x	-5
x	x^2	$-5x$
1	x	-5

$x^2 - 4x - 5 = (x - 5)(x + 1)$

1 mark
Factorise the quadratic.

$$\frac{3(x-1)}{(x-5)(x+1)} - \frac{2(x+1)}{(x-5)(x+1)}$$

1 mark
Use a common denominator.

$$= \frac{3x - 3 - 2x - 2}{(x-5)(x+1)}$$

$$= \frac{x-5}{(x-5)(x+1)}$$

1 mark
Simplify the numerator to x−5.

$$= \frac{1}{x+1}$$

1 mark
Cancel the common factor.

Original restrictions: $x \neq 5$ and $x \neq -1$.

(Total for Question 18 is 4 marks)